Calibration-Free Post-Estimation Fusion of Dual-View 3D Human Pose Streams: Robustness, Front-End Generalization, and Multi-View Extension

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Abstract

Monocular three-dimensional (3D) human pose estimates remain vulnerable to depth ambiguity, self-occlusion, and temporal jitter. A second camera can provide complementary evidence, but exploiting it through calibrated reconstruction is difficult when cameras follow a distant, fast-moving athlete or are installed temporarily. We present CanonFuse3D, a calibration-free post-estimation framework that fuses two independently estimated, approximately synchronized monocular 3D pose streams without requiring images, 2D observations, or camera parameters at the fusion stage. Human-centric canonicalization and sequence-level alignment expose cross-view agreement and disagreement; a shared joint encoder, bidirectional cross-view attention,

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and temporal refinement then drive joint-frame gating with residual correction. We evaluate the framework quantitatively on a synthetic Unity dataset with avatar ground truth and the real-world Ski-PTZ-Pose dataset with a constructed reference, and qualitatively on author-collected dual-view recordings. Relative to canonical averaging, CanonFuse3D reduces the mean per-joint position error by 44.5% on Unity and 48.6% on Ski-PTZ-Pose, while comparisons with learned and geometric references reveal metric- and input-dependent trade-offs rather than universal superiority. Controlled studies characterize the useful range across camera separations, small timing and sampling-rate errors, and pose or image occlusion, including failures under severe corruption. The same architecture supports alternative pose estimators through lightweight adaptation and additional cameras through pairwise composition.

Keywords: 3D human pose estimation, dual-view fusion, post-estimation pose fusion, calibration-free refinement, human-centric representation, front-end generalization

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1. Introduction

Monocular three-dimensional (3D) human pose estimation spans temporal lifting of 2D keypoints [1, 2, 3], pretrained motion representations [4], video-based body models [5], and foundation-scale full-body estimators [6]. These methods support sports motion analysis, but their outputs retain depth ambiguity, self-occlusion, viewpoint-dependent errors, and temporal jitter. Skiing combines rapid movement, distant observation, and continuous orientation changes, making these limitations particularly relevant [7, 8]. Complementary viewpoints can reveal a limb hidden in one camera and provide depth estimates with different errors.

Exploiting a second view usually requires information beyond the two monocular outputs. Binocular methods use stereo keypoints or jointly refine 2D and 3D poses through geometric constraints [9, 10]; calibrated multi-view methods triangulate detections or exchange image features under known geometry [11, 12, 13, 14]. Temporary placement and pan–tilt–zoom operation make calibration harder to maintain in skiing. Unknown-calibration methods relax that requirement: MetaPose initializes from monocular 3D estimates but retains 2D evidence when refining pose and camera variables [15]. The

relevant distinction is therefore what information cross-view refinement requires, rather than whether it reuses a monocular estimator.

We study post-estimation fusion when only two independently predicted, approximately synchronized 3D sequences on a common joint set are available to the fusion stage. The objective is a more accurate and temporally stable canonical sequence without supplying that stage with images, 2D detections, intrinsics, or extrinsics. These restrictions do not apply to the upstream estimators that process RGB frames. The output is human-centric rather than a recovered world-coordinate trajectory, and additional cameras are considered separately through pairwise composition. This interface allows an existing monocular pipeline to incorporate complementary predictions without introducing calibrated image-space reconstruction.

CanonFuse3D [16] (Figure 1) canonicalizes the streams using body landmarks and aligns them by sequence-level similarity transforms. Features describing pose agreement, disagreement, motion, and alignment feed a shared encoder, bidirectional cross-view attention, and temporal refinement. A joint-frame gate combines view-specific candidates with a learned residual correction. This article builds on the architecture of [16] and focuses on its robustness, front-end generalization, and extension to additional views. Keeping the upstream estimator separate allows the architecture to accept alternative estimators on the common joint set, although transfer of trained fusion weights requires experimental validation.

We evaluate the framework on three skiing datasets: a synthetic Unity dataset with avatar ground truth, the public Ski-PTZ-Pose dataset with a constructed canonical reference, and author-collected indoor and outdoor recordings used for qualitative deployment. Beyond clean-input accuracy, controlled studies examine how the same model behaves when the camera separation, timing, or sampling rate changes, when pose evidence or image regions are missing, when the upstream estimator is replaced, and when more than two cameras are available.

The main contributions are:

- A complete formulation of the human-centric fusion framework for calibration-free post-estimation dual-view 3D pose fusion, including its coordinate conventions, sequence-level alignment, and training objective, together with a controlled ablation of its reliability gate and residual correction.

- Quantitative evaluation on two skiing datasets with explicit reference

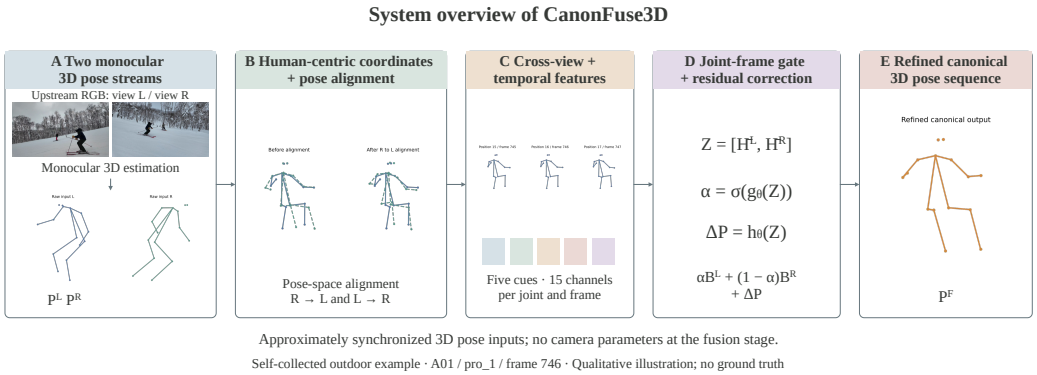


Figure 1: System overview of CanonFuse3D. (A) Two monocular 3D pose streams are estimated independently from the two views; (B) they are expressed in human-centric coordinates and aligned to each other in pose space; (C) cross-view and temporal features are constructed from five three-dimensional cues; (D) a joint-frame reliability gate and a residual head fuse the two view-specific candidates; (E) the output is a refined canonical 3D pose sequence. No camera parameters are used at any stage. The frames and skeletons are a qualitative example from the author-collected outdoor recordings and carry no ground truth.

regimes, separating canonical-space comparisons from an independent native-output audit and distinguishing pose-only baselines, published-method adaptations, and additional-input geometric references, complemented by qualitative deployment examples.

- A systematic robustness analysis covering camera separation, timing and sampling-rate errors, and both pose-level and image-level occlusion, which identifies the useful operating range and its failure cases.

2. Related Work

2.1. Monocular 3D Human Pose Estimation

Two-stage pipelines lift detected 2D trajectories to 3D using temporal convolutions (VideoPose3D [1]), spatial–temporal transformers (PoseFormer [2]), multiple hypotheses (MHFormer [3]), or pretrained motion representations (MotionBERT [4]). Separating lifting from detection permits matched-track comparisons, as in our alternative-front-end study, but errors in the 2D observations still propagate to 3D. The shared keypoint interface does not make the recovered streams interchangeable: estimator-specific limb

bias and temporal statistics can remain after coordinate alignment. Our controlled comparison therefore distinguishes the architecture's ability to accept a front end from the transferability of its trained weights.

Visual body estimators operate directly on images or video. SAM 3D Body recovers body, hand, and foot pose and shape from an image [6], while VIBE combines video features with an adversarial motion prior [5]. Both can supply pose streams to a downstream refinement stage. Keeping that stage separate avoids replacing the upstream estimator, but also limits refinement to information retained in its predicted poses.

Sports impose additional distribution shifts through rapid, atypical motion and self-occlusion. Rhodin et al. introduced Ski-Pose PTZ-Camera and used multi-view consistency to supervise monocular prediction [7]; SportsCap instead uses sub-motion priors for monocular sports videos [8]. Both remain monocular at inference. Our setting assumes two simultaneously available streams and studies how their potentially different errors can be reconciled after estimation.

2.2. Dual-view and Multi-view 3D Pose Estimation

Binocular methods exploit geometric redundancy within reconstruction. RSB-Pose estimates stereo keypoints with disparity-indexed features and refines poses with a pretrained transformer [9]. Dual-Diffusion jointly denoises binocular 2D and 3D poses using geometry and baseline-dependent normalization [10]. Their image/2D and stereo-geometry interfaces differ from refinement supplied only with predicted 3D streams.

Markerless multi-view estimation faces continuing challenges in occlusion, camera variation, supervision, temporal consistency, view selection, and computational cost [17]. Learnable Triangulation combines views through differentiable algebraic or volumetric triangulation [11]. Epipolar Transformers transfer image features along calibrated epipolar lines [12], while MVGFormer alternates appearance processing with geometric triangulation of refined 2D poses [13]. AdaFuse estimates heatmap correspondences and camera-specific weights to limit unreliable-view contributions [14]. These methods reconcile observations inside an image-, feature-, or geometry-based estimator.

MetaPose is closer to post-estimation fusion: it initializes pose and camera variables from monocular 3D predictions, then refines them with a learned optimizer and 2D uncertainty, without supplied calibration or multi-view 3D supervision [15]. Its 2D and projection terms remain additional inputs in our

labeled two-view adaptation. FLEX predicts view-invariant skeletal rotations and bone lengths from multiple streams without supplied extrinsics [18]. Applying general multi-view methods to two cameras does not remove their observation requirements. Our distinction is the restricted 3D-only interface, not the invention of two-view fusion. Accordingly, calibration-free methods should be compared by their required observations and recovered representation, rather than grouped solely by whether external camera parameters are supplied. The benchmark retains these distinctions instead of presenting additional-input methods as identical-interface replacements.

Takahashi et al. [19] use detected 2D joints to jointly estimate extrinsics, temporal offsets, and 3D poses from unsynchronized, uncalibrated cameras. CanonFuse3D instead assumes approximate synchronization and measures its residual timing tolerance experimentally.

2.3. Pose-Space Refinement, Reliability, and Generalization

STRIDE fits a motion prior to each video through model-agnostic test-time training, refining noisy 3D estimates under occlusion [20]. Its focus is temporal refinement of one video rather than reconciliation of synchronized views. Reliability weighting also operates within Learnable Triangulation and AdaFuse [11, 14]. Here the gate acts after monocular estimation, weighting aligned 3D candidates at the joint–frame level. Upstream confidence, feature quality, and residual pose-space reliability are therefore distinct signals. A confidence score attached to an image observation does not directly characterize disagreement after pose alignment, and the learned gate must be evaluated against its actual candidates rather than interpreted as a raw-camera confidence.

Distribution adaptation addresses another deployment boundary. PoseAug differentiably varies posture, body size, viewpoint, and position to augment paired 2D–3D training poses [21]. AdaptPose generates target-like motions and cameras using target-domain 2D poses without target 3D labels or camera information [22]. Both adapt a pose estimator's training distribution. Our experiments instead keep the fusion architecture fixed, replace its upstream estimator, and compare frozen transfer with front-end-specific or mixed adaptation. The evaluation thus tests both the benefit of complementary 3D predictions and the conditions under which that benefit fails.

3. Proposed Method

3.1. Overview

CanonFuse3D [16] maps two independently estimated, approximately synchronized pose sequences $P^L, P^R \in \mathbb{R}^{T\times J\times 3}$ to a canonical sequence P^F of the same shape, where T and J denote frames and joints. Images and 2D observations remain upstream; the fusion module neither receives camera parameters nor reconstructs a world-coordinate trajectory. Figure 2 groups the pipeline into canonicalization and alignment (stages 1–3), five-cue feature construction (stage 4), and cross-view temporal refinement with gated residual fusion (stages 5–8).

3.2. Human-Centric Landmark Canonicalization

Independent monocular estimates can differ in global position and orientation. We express each stream relative to a fixed reference pose A using the hip midpoint, hip direction, and neck direction. Let h_L, h_R, n denote the left hip, right hip, and neck, and let $\mathrm{nrm}(v) = v/(\|v\|_2+\epsilon)$. The origin and body axes are

$$c = \tfrac{1}{2}(A_{h_L} + A_{h_R}), \quad e_x = \mathrm{nrm}(A_{h_R} - A_{h_L}),$$
$$e_y = \mathrm{nrm}(A_n - c), \quad e_z = \mathrm{nrm}(e_x \times e_y).$$

An additional landmark-direction cue resolves the sign of e_z. With the axis matrix $R = [e_x, e_y, e_z]$, the transformed coordinates are

$$\tilde{P}_{t,j} = (P_{t,j} - c)R.$$

The left stream, right stream, and supervision reference use separate body references. Each origin and axis matrix remains fixed over the sequences to which it is applied; the representation is therefore not a per-frame root-relative pose. Skeleton scale is not normalized, and the constructed axes need not be orthogonal. In the evaluated protocol, the reference is shared within each batch. Its exact selection and landmark conventions are specified in Supplementary Section S9.

3.3. Pose-Level Cross-View Alignment

To reduce residual translation, orientation, and scale mismatch, we align each anchored stream to the other:

$$\tilde{P}^{L\to R} = \mathcal{A}(\tilde{P}^L, \tilde{P}^R), \qquad \tilde{P}^{R\to L} = \mathcal{A}(\tilde{P}^R, \tilde{P}^L),$$

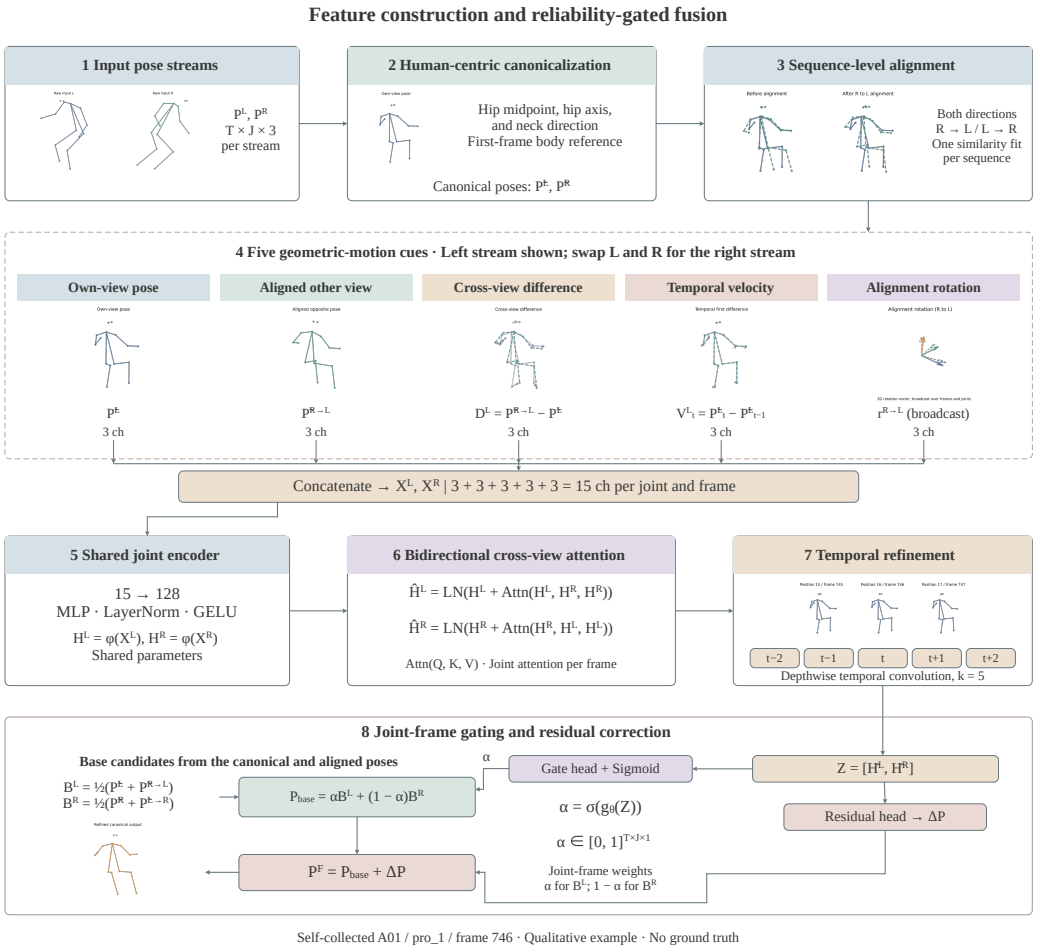


Figure 2: CanonFuse3D pipeline and feature construction. Stages 1–3 canonicalize and align the two pose streams; stage 4 concatenates five three-dimensional cues (own-view pose, aligned opposite-view pose, their difference, temporal velocity, and alignment rotation) into a 15-channel feature at every joint and frame; stages 5–8 encode the features, apply bidirectional cross-view attention and temporal refinement, and fuse the two base candidates B^L and B^R through a joint-frame gate α with a residual correction ΔP. The right stream uses the same construction with the view roles reversed.

where $\mathcal{A}$ estimates a sequence-level similarity transform.

For each sample, let $X, Y \in \mathbb{R}^{N\times 3}$ contain the source and destination joint coordinates in row-vector form, with $N = TJ$. We estimate one uniform scale

s, proper rotation Q, and translation d for the whole sequence:

$$(s^\star, Q^\star, d^\star) = \underset{s\geq 0,\ Q\in SO(3),\ d\in\mathbb{R}^3}{\arg\min} \frac{1}{N} \left\| sXQ^\top + \mathbf{1}d^\top - Y \right\|_F^2 .$$

We solve this unweighted least-squares problem by singular value decomposition with the proper-rotation correction of Umeyama [23], excluding reflections. Each direction uses one transform for all frames and joints in the sequence. These auxiliary pose-space transforms do not estimate camera extrinsics; they are also distinct from the per-frame reference alignment used only for evaluation.

The model requires dense, finite pose tensors and receives no joint-validity mask. Missing detections are represented by zero-filled raw poses, and the controlled pose-corruption study replaces or perturbs selected raw entries before canonicalization. All transformed entries participate equally in alignment and training, including corrupted or zero-filled entries; missing joints are neither interpolated nor omitted. Detection-failure flags are used only for reporting, and supervision references remain uncorrupted. Supplementary Section S10 gives the intervention rules.

We further compute cross-view residuals:

$$D^L = \tilde{P}^{R\to L} - \tilde{P}^L, \qquad D^R = \tilde{P}^{L\to R} - \tilde{P}^R.$$

These residuals encode disagreement between the two pose streams and provide cues about view-specific errors.

3.4. Geometric-Motion Feature Construction

Stage 4 of Figure 2 concatenates five three-dimensional cues at every joint and frame:

$$X^L = [\tilde{P}^L, \tilde{P}^{R\to L}, D^L, V^L, r^{R\to L}],$$
$$X^R = [\tilde{P}^R, \tilde{P}^{L\to R}, D^R, V^R, r^{L\to R}].$$

Velocity is the first temporal difference, $V_t^L = \tilde{P}_t^L - \tilde{P}_{t-1}^L$ with $V_1^L = 0$, and likewise for V^R. Each alignment rotation vector r is broadcast across all joints and frames. The resulting 15-dimensional feature describes the own-view estimate, aligned opposite estimate, disagreement, local motion, and sequence-level rotation. The two pose cues expose alternative configurations in one stream's pose space, while their residual measures disagreement explicitly. Velocity supplies local motion context, and the broadcast rotation

describes the global relation between streams. These cues inform candidate reliability; they are feature components, not separately trained prediction branches.

3.5. Joint Feature Encoding

A shared joint-wise encoder maps the 15-dimensional features to 128-dimensional hidden representations:

$$H^L = \phi(X^L), \qquad H^R = \phi(X^R).$$

The encoder ϕ is a two-layer multilayer perceptron, each layer followed by layer normalization and GELU activation, preserving the joint and frame indices. Weight sharing applies the same feature mapping to both views. Cross-view exchange occurs in the following attention stage rather than inside this joint-wise encoder.

3.6. Bidirectional Cross-View Attention

Within each frame, each view attends to the other view's J joint features:

$$\hat{H}^L = \mathrm{LN}\Big(H^L + \mathrm{Attn}(Q = H^L, K = H^R, V = H^R)\Big),$$
$$\hat{H}^R = \mathrm{LN}\Big(H^R + \mathrm{Attn}(Q = H^R, K = H^L, V = H^L)\Big).$$

The two multi-head attention directions have separate parameters. Residual connections and layer normalization retain the own-view features while incorporating cross-view evidence. Attention spans joints within a frame, allowing both corresponding and related joints to contribute. Temporal interaction is handled by the next module.

3.7. Temporal Refinement

For each joint, temporal refinement applies layer normalization, a depthwise one-dimensional convolution of kernel size 5, GELU, dropout, and a pointwise convolution:

$$\bar{H} = \hat{H} + \mathrm{TempConv}(\mathrm{LN}(\hat{H})).$$

The same module processes both streams, adding short-range motion context to their cross-view features. Its residual path preserves the input representation while learning a local correction. This temporal feature update precedes the gate and is distinct from smoothing the final output poses.

3.8. Reliability-Gated Fusion and Residual Correction

Concatenated features drive two two-layer perceptron heads with GELU activation:

$$Z = [\bar{H}^L, \bar{H}^R], \qquad \alpha = \sigma(g_\theta(Z)), \qquad \Delta P = h_\theta(Z).$$

Here σ is logistic and $\alpha \in [0,1]^{T\times J\times 1}$ weights two view-specific pose candidates:

$$B^L = \tfrac{1}{2}(\tilde{P}^L + \tilde{P}^{R\to L}), \qquad B^R = \tfrac{1}{2}(\tilde{P}^R + \tilde{P}^{L\to R}),$$
$$P^{base} = \alpha B^L + (1-\alpha)B^R, \quad P^F = P^{base} + \Delta P.$$

Because the streams have separate body references, this learned pose-space combination does not imply a shared physical frame. The gate is a learned reliability signal, not calibrated probabilistic uncertainty.

3.9. Training Objective

The model is trained with a supervised pose loss and four lightweight regularization terms:

$$\mathcal{L} = \mathcal{L}_{sup} + \lambda_{view}\mathcal{L}_{view} + \lambda_{bal}\mathcal{L}_{bal} + \lambda_{ent}\mathcal{L}_{ent} + \lambda_{temp}\mathcal{L}_{temp}.$$

The supervised term is the mean absolute error to the independently canonicalized reference $\tilde{G}$,

$$\mathcal{L}_{sup} = \frac{1}{3TJ}\sum_{t,j} \|P^F_{t,j} - \tilde{G}_{t,j}\|_1.$$

The view-reconstruction term preserves the discrepancy between the two canonicalized inputs. With $\delta = \tilde{P}^L - \tilde{P}^R$, the fused pose and gate reconstruct the two inputs as $\hat{P}^L = P^F + (1-\alpha)\delta$ and $\hat{P}^R = P^F - \alpha\delta$, and

$$\mathcal{L}_{view} = \frac{1}{2}\left(\frac{1}{3TJ}\sum_{t,j}\|\hat{P}^L_{t,j} - \tilde{P}^L_{t,j}\|_1 + \frac{1}{3TJ}\sum_{t,j}\|\hat{P}^R_{t,j} - \tilde{P}^R_{t,j}\|_1\right).$$

This term constrains the gate and fused estimate without treating the sequence-level similarity transforms as physical camera geometry. Two gate penalties avoid degenerate fusion: the balance term $\mathcal{L}_{bal} = |\bar{\alpha} - 0.5|$, where $\bar{\alpha}$ is the mean gate over all joints and frames, discourages collapse to a single view, and the negative-entropy term

$$\mathcal{L}_{ent} = \frac{1}{TJ}\sum_{t,j}[\alpha_{t,j}\log\alpha_{t,j} + (1-\alpha_{t,j})\log(1-\alpha_{t,j})]$$

discourages overconfident gates during training. Finally, the temporal term penalizes second-order variation of the fused trajectory,

$$\mathcal{L}_{temp} = \frac{1}{(T-2)J} \sum_{t=2}^{T-1} \sum_{j} \| P^F_{t+1,j} - 2P^F_{t,j} + P^F_{t-1,j} \|_2,$$

encouraging stable motion without removing the supervised accuracy objective. The weights are $\lambda_{view} = 0.05$, $\lambda_{bal} = 0.02$, $\lambda_{ent} = 0.005$, and $\lambda_{temp} = 0.01$. All sums are additionally averaged over the mini-batch. No term uses an input-visibility mask to omit or reweight missing joints: an available reference still supervises the output at every joint, while a zero-filled input remains part of the view-reconstruction target.

4. Experimental Protocol

4.1. Experimental Setup

Upstream estimators process the views independently. CanonFuse3D and pose-only controls receive their 3D sequences; geometric controls additionally receive estimated 2D joints and supplied calibration. Within each dataset and coordinate protocol, comparisons use matched sequences and joint conventions.

4.2. Data Sources and Annotation

We use three skiing datasets with complementary roles: two for quantitative evaluation and one for qualitative deployment. Table 1 summarizes their content, scale, and reference regimes; Figure 3 shows paired-view examples.

Unity synthetic dataset. The Unity scene [24] contains one female and one male avatar performing steady skiing, braking, falls, and transitions. Table 1 gives the action and camera counts; azimuths are spaced at 10-degree intervals. Each sample comprises one camera pair. Avatar 3D ground truth is independent of the monocular predictions.

Ski-Pose PTZ-Camera dataset. Ski-PTZ-Pose records competitive giant-slalom skiers with synchronized pan-tilt-zoom cameras and provides 3D poses, projected 2D annotations, and per-frame calibration [7, 25]. We align view-specific 3D annotations to a selected camera using similarity transforms fitted on an initial segment. The coordinate-wise median of the aligned

Table 1: Data sources. Split counts are train / validation / test camera pairs; the Unity split is action-disjoint.

	Unity synthetic	Ski-PTZ-Pose	Author-collected videos
Cameras	180 synchronized views per action (5 layers × 36 azimuths)	6 synchronized PTZ cameras	2 per recording
Sequences	24 (2 avatars × 12 actions), 20–65 frames	12 runs, about 10K images	6 recordings, 360–2,609 frames per view
Reference	Avatar skeleton, 15 joints	Constructed canonical reference, 13 joints	None
Split	193,320 / 128,880 / 64,440	150 / – / 30	Not used
Role	Training and all quantitative studies	Real-world evaluation	Qualitative only

annotations on the remaining frames defines the supervision and evaluation reference. Errors measure agreement with this constructed canonical reference, not independent physical 3D accuracy. Neither calibration nor annotation-derived transforms enter CanonFuse3D.

Author-collected qualitative videos. Six indoor and outdoor dual-view recordings illustrate deployment. Each view is processed independently. These videos have no independent metric 3D annotations and contribute neither training samples nor quantitative evaluation.

Joint convention. The common 13-joint evaluation set comprises the bilateral shoulders, elbows, hands, hips, knees, and ankles, plus the neck (indices 2–14 of the native 15-joint skeleton). The Unity rig and SAM3D additionally provide two eye landmarks, which help resolve facing direction during canonicalization; Ski-PTZ-Pose and the alternative lifting front ends do not provide these landmarks. The main comparison on both datasets and all cross-front-end comparisons use the common 13 joints. Earlier controlled native-input Unity studies use all 15 joints, whereas the matched RGB-occlusion and calibration study reports common-13 errors after all-15 canonicalization (Section 5.7).

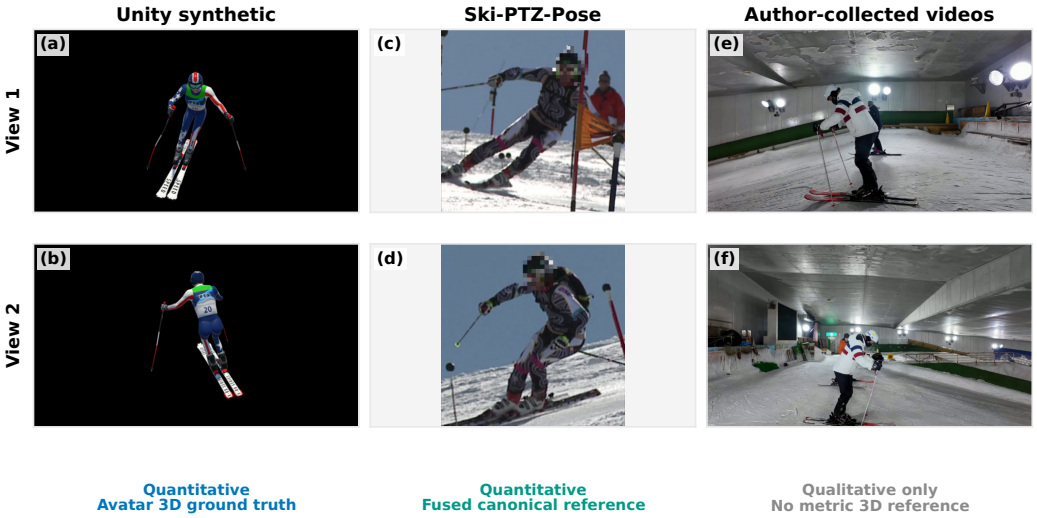


Figure 3: Representative synchronized dual-view examples from the three data sources. Columns show (a,b) Unity synthetic data, (c,d) Ski-PTZ-Pose, and (e,f) author-collected videos; rows show the two input views. Unity and Ski-PTZ-Pose support quantitative evaluation under the reference regimes stated in the text. The author-collected recordings have no independent metric 3D annotations and are used only for qualitative illustration. The Unity panels are foreground-cropped for legibility; the other panels retain their original field of view.

4.3. Implementation Details

Each camera-pair sample is uniformly resampled to 30 shared frames. The refinement model uses a shared two-layer joint encoder with hidden dimension 128, four-head bidirectional cross-view attention, a depthwise temporal convolution of kernel size 5, and two-layer gate and residual heads, and is trained with AdamW. Unity uses action-disjoint training, validation, and test splits; Ski-PTZ-Pose uses its predefined training and test partitions. Unity model selection uses validation error, while Ski-PTZ-Pose uses predeclared checkpoints because no independent validation partition is available. Test errors do not determine model selection. Complete architectures, optimization settings, split sizes, and checkpoint selection details are provided in Supplementary Section S9.

Pose-only controls include the single views, canonical averaging, sequence-aligned averaging [23], a 3D quality-weighted average, a temporal convolutional network (TCN) trained on the same data, and SmoothNet [26] adapted to canonical averaging. The quality heuristic and three-layer, non-

dilated TCN are study-specific controls. Temporal convolution has established sequence-modeling precedent [27], but our TCN does not reproduce that architecture. SmoothNet's architecture and adaptation settings are documented in the supplement.

Geometric controls comprise classical DLT [28], reprojection-filtered DLT, and DLT with learned residual correction. The latter two are study-specific extensions, not reproductions of published learned-triangulation systems. Their additional estimated 2D observations and calibration are labeled in the main table.

4.4. Evaluation Metrics

Errors use the coordinate convention of the corresponding evaluation protocol, without conversion to millimeters. Table 2 and the controlled canonical-space studies apply the stated body-coordinate transformation independently to predictions and references. Table 3 instead evaluates admissible native outputs before that transformation and removes only per-frame pelvis translation for position error; PA-MPJPE remains a separate evaluation-only shape alignment. The equations below use the canonical-space notation; the native audit substitutes its declared prediction and reference coordinates. Lower values indicate better agreement with the corresponding reference, and neither coordinate protocols nor dataset reference regimes are pooled.

4.4.1. Mean per-joint position error

Mean per-joint position error (MPJPE) averages Euclidean distances between predicted and reference joint positions:

$$E_{\mathrm{MPJPE}} = \frac{1}{TJ} \sum_{t=1}^{T} \sum_{j=1}^{J} \| P_{t,j}^{F} - \tilde{G}_{t,j} \|_2 .$$

Dataset-level results aggregate all evaluated joint-frame points. No additional reference alignment is applied for this metric.

4.4.2. Procrustes-aligned position error

Procrustes-aligned MPJPE (PA-MPJPE) evaluates pose shape after fitting a least-squares similarity transform from prediction to reference independently in each frame:

$$E_{\mathrm{PA}} = \frac{1}{TJ} \sum_{t=1}^{T} \sum_{j=1}^{J} \left\| s_t P_{t,j}^{F} Q_t^{\top} + d_t - \tilde{G}_{t,j} \right\|_2 ,$$

where, for each frame t,

$$(s_t, Q_t, d_t) = \underset{s\geq 0,\, Q\in SO(3),\, d\in\mathbb{R}^3}{\arg\min} \sum_{j=1}^{J} \left\| sP_{t,j}^{F} Q^{\top} + d - \tilde{G}_{t,j} \right\|_2^2 .$$

The fit allows translation, a nonnegative uniform scale, and a proper rotation, with no reflection. Each reported joint subset is aligned separately. Collapsed predictions are mapped to the reference centroid and retained in the average. This reference fit is used only for PA-MPJPE and does not change model inputs, selection, or other metrics.

4.4.3. Acceleration error

Acceleration error measures agreement between second-order pose differences:

$$E_{\mathrm{acc}} = \frac{1}{(T-2)J} \sum_{t=2}^{T-1} \sum_{j=1}^{J} \|\Delta^2 P_{t,j}^{F} - \Delta^2 \tilde{G}_{t,j}\|_2,$$

where $\Delta^2 P_{t,j} = P_{t+1,j} - 2P_{t,j} + P_{t-1,j}$. It measures temporal agreement at the evaluated sampling interval, in coordinate units per sampled frame squared.

4.5. Robustness and Generalization Protocols

The robustness studies use the frozen Unity checkpoint and the complete held-out split; additional training is restricted to the explicitly identified gate ablations and front-end adaptation experiments. Camera separation is analyzed in six 30-degree bins of circular azimuth difference between the two cameras. Timing robustness uses seven right-stream offsets from -2 to $+2$ frames, including half-frame conditions, and seven sampling-rate errors from -2% to $+2\%$; an auxiliary diagnostic additionally estimates larger offsets of up to five frames. Pose-level corruption crosses three affected-view modes, three patterns, and five ratios, giving 45 conditions, and image-level occlusion applies the same view modes and patterns at two ratios to the RGB frames before the native pose estimator, giving 18 conditions; the two ratios parameterize different interventions and do not imply equal physical severity. Front-end generalization replaces SAM3D by three lifting estimators applied to the same ground-truth 2D tracks, isolating the lifting stage, followed by frozen transfer and target-specific or mixed adaptation. Finally, the multi-view extension composes the frozen dual-view model over all camera pairs of nested one- to four-camera groups. Full intervention and aggregation rules are given in Supplementary Section S10.

A separate matched-input study compares five frozen methods on clean images and random ratio-0.5/1.0 masks applied to the left, right, or both views. Paired estimated 2D and 3D observations come from the same SAM3D detections. The geometry-assisted methods additionally receive calibration, with right-camera extrinsic rotation perturbations of 0°, 1°, and 3°. All seven conditions retain the full test cohort, including detection failures (Section 5.7).

5. Results

5.1. Primary performance and cross-dataset evidence

Table 2 compares CanonFuse3D with the pose-only fusion baselines, adapted published methods, and the separately labeled geometric group on the common 13-joint subset of both datasets after the shared body-coordinate normalization. On all 64,440 held-out Unity camera-pair sequences, CanonFuse3D reduces MPJPE from 0.2792 for canonical averaging to 0.1551 (44.5%). Within this canonical-space protocol, the right row has larger MPJPE than the left row (0.4617 versus 0.1624). This gap cannot by itself be attributed to worse native front-end predictions because the streams pass through separate batch-referenced coordinate transforms; the native-output audit below evaluates the upstream predictions in declared frames. Aligning the right stream to the left before averaging reduces the canonical-space error to 0.1622, and the quality-weighted average brings no further gain. CanonFuse3D improves on the best of these non-learned methods in MPJPE (0.1551 versus 0.1622), PA-MPJPE (0.0637 versus 0.0667), and acceleration error (0.0297 versus 0.0308), so the learned fusion adds accuracy and temporal stability beyond what pose-space alignment alone provides. It is also more accurate than the temporal convolutional baseline trained on the same data (0.1610 MPJPE, 0.0690 PA-MPJPE, 0.0339 acceleration error). SmoothNet applied to canonical averaging gives the lowest acceleration error in the group (0.0113) but leaves the position error at 0.2925, so temporal smoothness and positional accuracy must be read separately. On all 15 Unity joints, the corresponding reduction over canonical averaging is 46.3% (0.1541 versus 0.2868; Supplementary Table S1).

On the 30 fixed Ski-PTZ-Pose test sequences, the two views disagree far more (0.3552 and 0.4462), and pose-space alignment alone does not resolve the disagreement (0.3537). CanonFuse3D reduces MPJPE to 0.1794, a 48.6% reduction over canonical averaging, with PA-MPJPE 0.1269 and acceleration

error 0.0311. On this small real-world index the temporal convolutional baseline reaches lower MPJPE and PA-MPJPE (0.1581 and 0.1144) at a higher acceleration error (0.0411), and SmoothNet again gives the lowest acceleration error (0.0151) with a position error close to averaging (0.3227). These values measure agreement with the constructed annotation reference rather than independent physical 3D accuracy, and the 30-sequence index provides limited resolution between the two learned models.

The geometric group quantifies what calibration buys. With the estimated 2D joints and the supplied calibration, plain DLT is less accurate than CanonFuse3D on both datasets (0.1983 and 0.3984), reprojection gating does not improve it, and only a learned residual correction on top of DLT reaches lower MPJPE (0.0964 and 0.1643), at the cost of requiring 2D observations and camera calibration that the pose-only setting excludes. This group is therefore not a common-input comparison with CanonFuse3D. On Ski-PTZ-Pose the geometric predictions are first centered at their own per-frame pelvis, consistent with the root-relative annotations; this preprocessing fits nothing to the test reference.

The published adaptations give a metric-dependent comparison under their stated contracts. Avg. + STRIDE [20] remains close to canonical averaging, with Unity MPJPE 0.2799, PA-MPJPE 0.1174, and acceleration error 0.0316; its corresponding Ski-PTZ-Pose values are 0.3483, 0.1861, and 0.0387. The official-protocol DeciWatch adaptation [29] reaches 0.1134, 0.0695, and 0.0132 on Unity and 0.2239, 0.1549, and 0.0168 on Ski-PTZ-Pose. It has lower Unity MPJPE and acceleration error than CanonFuse3D but higher PA-MPJPE; on Ski-PTZ-Pose, CanonFuse3D has lower MPJPE and PA-MPJPE while DeciWatch has lower acceleration error. MetaPose [15], which additionally uses estimated 2D joints, reaches 0.2655, 0.1089, and 0.0400 on Unity and 0.4253, 0.1425, and 0.0634 on Ski-PTZ-Pose. These adapted methods differ in input, supervision, and temporal-recovery task, so the table reports their outcomes without treating the lowest cell in each column as a common-input winner.

Table 3 instead evaluates outputs before the shared body-coordinate normalization. On Unity, the pooled native SAM3D streams have MPJPE/PA-MPJPE/acceleration error 0.1019/0.0732/0.0369. The native left stream is slightly more accurate than the right (0.0967 versus 0.1070 MPJPE), while sequence alignment and quality weighting reduce PA-MPJPE to 0.0666 and 0.0660. Among the published-method adaptations, per-view STRIDE [20] reaches 0.1072/0.0776/0.0357, per-view DeciWatch [29] reaches

Table 2: Comparison after the shared body-coordinate normalization on the common 13-joint subset. Unity uses all 64,440 held-out camera-pair sequences; Ski-PTZ-Pose uses all 30 fixed test sequences. Published adaptations are separated because their input and task budgets differ: STRIDE refines the dense canonical average, DeciWatch recovers 30 frames from four observed source frames, and MetaPose additionally receives estimated 2D joints. Bold marks the lowest value only within the original study-specific input blocks and does not imply an overall ranking.

	Unity			Ski-PTZ-Pose		
Method	MPJPE ↓	PA-MPJPE ↓	Accel. ↓	MPJPE ↓	PA-MPJPE ↓	Accel. ↓
3D pose inputs only						
Left canonical	0.1624	0.0711	0.0355	0.3552	0.1864	0.0531
Right canonical	0.4617	0.0759	0.0452	0.4462	0.1754	0.0455
Canonical average	0.2792	0.1162	0.0322	0.3487	0.1877	0.0392
Aligned average [23]	0.1622	0.0667	0.0308	0.3537	0.1718	0.0381
3D quality-weighted fusion	0.2767	0.1152	0.0312	0.3499	0.1875	0.0385
Temporal fusion (local TCN) [27]	0.1610	0.0690	0.0339	**0.1581**	**0.1144**	0.0411
Avg. + SmoothNet [26]	0.2925	0.1860	**0.0113**	0.3227	0.1924	**0.0151**
Published temporal adaptations of the canonical average						
Avg. + STRIDE (adapted) [20]	0.2799	0.1174	0.0316	0.3483	0.1861	0.0387
DeciWatch (official-protocol adaptation) [29]	0.1134	0.0695	0.0132	0.2239	0.1549	0.0168
Proposed pose-only fusion						
CanonFuse3D (ours)	**0.1551**	**0.0637**	0.0297	0.1794	0.1269	0.0311
Additional estimated 2D joints without camera calibration						
MetaPose (adapted) [15]	0.2655	0.1089	0.0400	0.4253	0.1425	0.0634
Additional estimated 2D joints and camera calibration						
Calibrated DLT [28]	0.1983	0.1146	0.0376	0.3984	0.2219	0.0816
Reprojection-gated DLT [28]	0.2088	0.1196	**0.0295**	0.3996	0.2234	0.0846
DLT-residual MLP [28]	**0.0964**	**0.0535**	0.0403	**0.1643**	**0.1172**	**0.0598**

0.6315/0.3033/0.0119, and MetaPose [15] reaches 0.2171/0.1088/0.0382. DeciWatch therefore smooths its per-view inputs but incurs large position error under this native-view application; this row is distinct from the canonical-average recovery protocol in Table 2.

Every method in the Unity-only Table 3 is scored directly in its declared output frame without the CanonFuse3D body-coordinate transform. Monocular and temporal-refinement results pool independently scored left and right camera streams; MetaPose remains in its restored left-camera frame, and calibrated geometry remains in its supplied world frame. Ski-PTZ-Pose is omitted from this table because its constructed annotation reference has no independently supplied mapping into these frames; native-space values are withheld rather than created by fitting a transform to test targets. The literal unaligned average is likewise retained only as a coordinate-mixing diagnostic. CanonFuse3D appears only in the canonical-space table; no native score is inferred through an unverified inverse. Because the protocols differ

Table 3: Independent Unity comparison before the shared body-coordinate normalization. All listed methods operate on their native outputs and do not use CanonFuse3D's canonical transform. Position error removes only per-frame pelvis translation; Procrustes alignment is a separate reflection-free diagnostic. A double dash denotes that no target-independent common coordinate frame or admissible output gauge was available.

Method	MPJPE ↓	PA-MPJPE ↓	Accel. ↓
Native monocular front ends			
SAM 3D Body [6]	0.1019	0.0732	0.0369
MotionBERT [4]	0.2553	0.1150	0.0090
PoseFormer [2]	0.2075	0.1371	0.0068
VideoPose3D [1]	0.1583	0.1028	0.0076
Pre-Canon controls			
Left native	0.0967	0.0708	0.0333
Right native	0.1070	0.0756	0.0406
Unaligned native average	–	–	–
Sequence-aligned average [23]	0.0970	0.0666	0.0285
Aligned quality-weighted fusion	0.0960	0.0660	0.0276
Aligned average + SmoothNet [26]	–	–	–
Published external methods			
STRIDE (per-view adaptation) [20]	0.1072	0.0776	0.0357
DeciWatch (per-view adaptation) [29]	0.6315	0.3033	0.0119
MetaPose (adapted) [15]	0.2171	0.1088	0.0382
Calibrated geometry			
Calibrated DLT [28]	0.1705	0.1072	0.0614
Reprojection-gated DLT [28]	0.1711	0.1098	0.0520
DLT-residual MLP [28]	–	–	–

by whether the shared normalization is applied, methods must not be ranked across the two tables.

5.2. Reliability-gate behavior

Table 4 crosses dynamic versus fixed-0.5 gating with presence versus absence of the output residual. Three learned configurations are independently retrained for 100 epochs on the same Unity partition, seed 42, optimization, and validation-selection rules. The fourth has no trainable output path. All use the complete test index and batch size 256 (Supplementary Section S12). This full-model control is distinct from the archived benchmark checkpoint in Table 2; the subsequent experiment instead intervenes on that frozen checkpoint.

With the output residual retained, learned gating lowers MPJPE from

Table 4: Controlled gate/output-residual ablation on Unity (common-13, canonical coordinates). Each learned configuration is trained from scratch with the same seed, split, budget, and validation selection rule. This independent experiment does not replace the archived main-benchmark model. Lower is better.

Gate	Output residual	MPJPE	PA-MPJPE	Accel.
Dynamic	Learned	**0.14639**	**0.06734**	0.02953
Fixed 0.5	Learned	0.17279	0.07011	0.03012
Dynamic	None	0.16334	0.07337	0.02929
Fixed 0.5	None (deterministic)	0.27997	0.11486	**0.02513**

0.17279 to 0.14639 (15.3%) relative to fixed weights. With dynamic gating retained, the output residual lowers MPJPE from 0.16334 to 0.14639 (10.4%). The full model also has the lowest PA-MPJPE. These single-run comparisons support complementary contributions to position accuracy under this protocol, not a claim of seed-independent gains. Removing the residual slightly lowers acceleration error, and the deterministic configuration is smoothest but has the largest position error; temporal smoothness alone does not establish reconstruction quality.

In the frozen benchmark model, each gated candidate already combines both aligned raw views, so $\alpha = 1$ or 0 does not select an isolated camera. On all 15 Unity joints, correlation between α and right-minus-left candidate error is 0.651; the larger weight selects the lower-error candidate in 91.7% of non-tied joint-frame comparisons. Table 5 and Fig. 4 report this complete-split audit. Mean left-candidate weight is 0.6804, indicating asymmetric preference, with small across-joint mean variation but joint-frame standard deviation 0.1122. These are diagnostics of aligned-candidate errors in evaluation coordinates, not probabilistic calibration or raw-camera reliability.

Table 5: Region-wise reliability-gate statistics on the complete Unity test split without masking.

Region	Joints	α	$1-\alpha$
Upper limbs	6	0.6879 ± 0.0132	0.3121
Lower limbs	6	0.6686 ± 0.0103	0.3314
All joints	15	0.6804 ± 0.0144	0.3196

Note. Values report the mean and population across-joint standard deviation of per-joint mean α, not the standard deviation across all joint-frame weights. Upper limbs are the six shoulder/elbow/wrist joints and lower limbs the six hip/knee/ankle joints in the filtered skeleton. The all-joint group additionally includes both eyes and the neck. Balance and entropy regularization do not enforce a mean of 0.5 or probabilistic calibration.

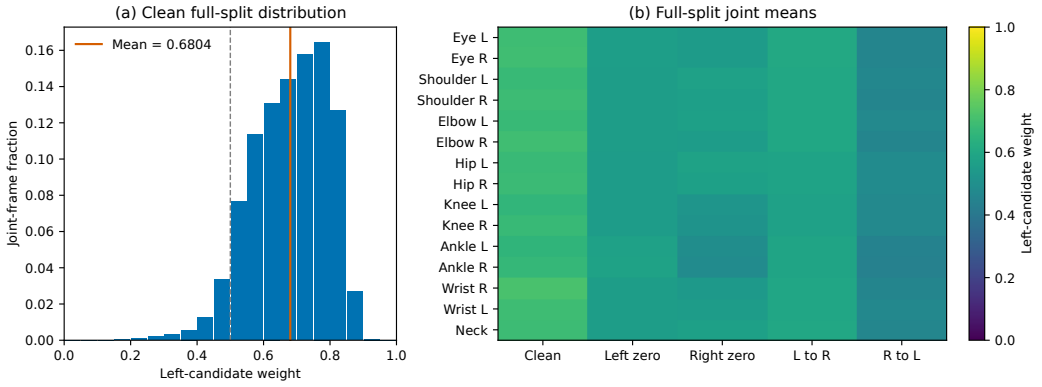


Figure 4: Frozen-checkpoint gate diagnostics on the complete Unity split (all 15 joints). (a) Clean joint-frame weight distribution. (b) Per-joint means under clean inputs, complete raw-pose zeroing of either stream, and midpoint switches of the zeroed stream. The latter conditions also disturb canonicalization and alignment; they do not isolate gate reliability.

Freezing this checkpoint and replacing only the gate by 0.5 raises clean common-13 MPJPE from 0.1551 to 0.2104 on Unity and from 0.1794 to 0.1842 on Ski-PTZ-Pose. These inference interventions differ from retraining a model to compensate for a fixed gate. Under complete raw-pose zeroing and midpoint switches, the original checkpoint retains only a small advantage over the fixed-gate intervention, and on Unity it is worse than canonical averaging in every such stress condition. Turning off its output residual reduces Unity error under these extreme perturbations but worsens clean error and all tested Ski conditions. Complete zeroing also disrupts the batch anchor and Sim3 alignment, so these results do not establish automatic recovery

of a missing view. All interventions, including unfavorable outcomes, are reported in Supplementary Table S9; fixed-index frame–joint examples are shown in Supplementary Fig. S3.

5.3. Camera-pair separation

The controlled studies in this and the following three subsections vary one capture condition at a time while keeping the checkpoint, held-out split, and evaluation path fixed. The camera-angle breakdown in Table 6 does not reveal a failure regime. CanonFuse3D MPJPE decreases from 0.1563 in the 0–30 degree bin to 0.1528 around 120–150 degrees, while the gain over canonical averaging stays between 45.9% and 46.6%. Wider azimuth separation therefore supplies slightly more complementary evidence in this synthetic setup, but the narrow range cautions against claiming a strong optimum. The bins contain 8,640–12,600 mutually exclusive test camera-pair sequences and together cover the complete 64,440-sequence test split.

To account for the four action-specific observations associated with each camera geometry, we collapse results by unordered camera pair before inference. Every angle bin retains a positive fusion gain under paired Wilcoxon testing (all Holm-adjusted $p < 0.001$; rank-biserial correlation 0.994–1.000). The across-bin Kruskal–Wallis test is statistically significant ($p = 0.025$) but has a negligible effect size ($\epsilon^2 = 0.00049$). Of 15 Holm-adjusted post-hoc comparisons, only 90–120 versus 150–180 degrees remains significant ($p = 0.0064$), again with a small rank-biserial effect ($r_{rb} = 0.053$). Thus, the fusion benefit is reliable within every tested bin, whereas camera separation explains little of its magnitude. Detailed paired-test statistics are provided in Supplementary Table S2; Supplementary Fig. S1 shows the corresponding absolute-error curves.

Table 6: Accuracy and camera-pair bootstrap intervals by azimuth separation. Positive gain favors CanonFuse3D.

Angle (deg)	Pairs	Canon. avg.	Fused	Mean gain	95% CI
0–30	2,160	0.2891	0.1563	0.1328	[0.1313, 0.1343]
30–60	2,700	0.2888	0.1554	0.1334	[0.1321, 0.1347]
60–90	2,700	0.2876	0.1544	0.1332	[0.1318, 0.1345]
90–120	2,700	0.2865	0.1529	0.1337	[0.1324, 0.1349]
120–150	2,700	0.2852	0.1528	0.1325	[0.1314, 0.1336]
150–180	3,150	0.2844	0.1532	0.1312	[0.1303, 0.1322]

Note. Each camera pair contributes four action-specific sequences; the bins cover all 64,440 test sequences. MPJPE and absolute gain use dataset coordinate units. Intervals resample camera pairs. Detailed paired tests and effect sizes appear in Supplementary Table S2.

5.4. Temporal offset and sampling-rate drift

Figure 5 summarizes two synchronization stress tests as relative MPJPE changes from each method's unperturbed value. Supplementary Table S3 provides the absolute errors and gate diagnostics. At zero perturbation, all-15 MPJPE is 0.1541. Across ± 2 frame offsets, it ranges from 0.1537 to 0.1555; the worst case, a two-frame positive offset, is only 0.91% above zero. Gate-error correlation stays between 0.646 and 0.650 and view-preference accuracy remains above 91.4%. The modest variation suggests that sequence-level alignment and the five-frame temporal refiner tolerate short local disagreement, but this experiment does not establish robustness to long delays or unsynchronized action phases.

Clock-rate perturbations are even smaller. At +2%, MPJPE increases by 0.23%; negative perturbations up to -2% change MPJPE by less than 0.02%. Acceleration error is not monotonic because temporal resampling smooths as well as misaligns trajectories. Consequently, MPJPE is the cleaner measure of rate-drift sensitivity in this controlled test.

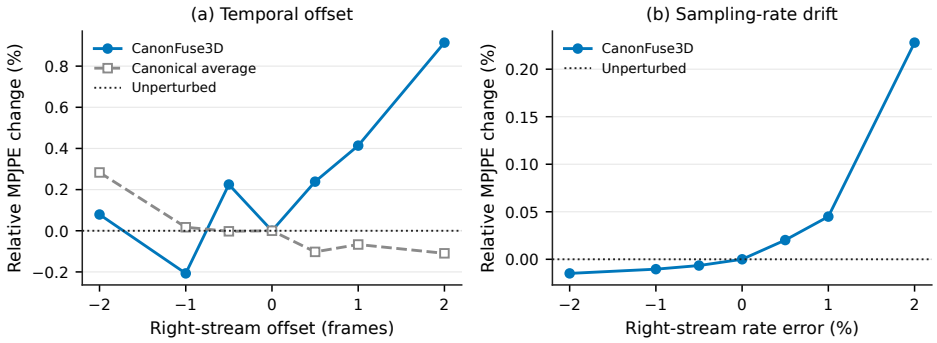


Figure 5: Relative MPJPE change under right-stream temporal offset and sampling-rate error on the complete held-out split. Each curve is normalized by its own method-specific unperturbed MPJPE; positive values indicate increased error. Dotted horizontal lines mark zero change.

5.5. Missing-pose evidence

Figure 6 reports all 45 conditions of the complete held-out-split missing-pose study; Supplementary Table S4 gives exact errors and gains at the representative ratios 0.5 and 1.0. Corrupted entries use a fixed mixture of zero replacement (35%) and Gaussian perturbation (65%, standard deviation 0.08 in raw input coordinate units, before canonicalization). For random masking, the ratio is the independent joint-frame probability; in the both-view condition, masks and corruption noise are sampled separately for the two streams. Distal-subset masking applies that probability only to the same eight filtered joint indices used by the executable masking protocol. Temporal masking selects the stated fraction of joints and corrupts one independently positioned contiguous 10-frame interval for each selected joint within every 30-frame sequence; its ratio therefore denotes selected joints rather than the global fraction of tensor entries.

Single-view corruption tests whether the complementary stream can rescue degraded evidence, whereas both-view corruption probes the method when redundancy is removed. The curves compare fused and canonical-average MPJPE, making explicit where learned fusion retains an advantage and where both methods fail together. At ratio 0.5, single-view temporal masking retains gains of 37.0% for left corruption and 37.9% for right corruption; distal masking retains 28.0% and 31.5%, respectively. The advantage narrows for unstructured random corruption. Most importantly, when random corruption affects both streams at ratio 0.5, fusion is 3.3% worse than

canonical averaging, whereas bilateral temporal and distal corruption still retain gains of 26.9% and 12.5%. Thus, the checkpoint can exploit a clean complementary stream and structured missingness, but dense independent corruption in both inputs lies outside its reliable regime.

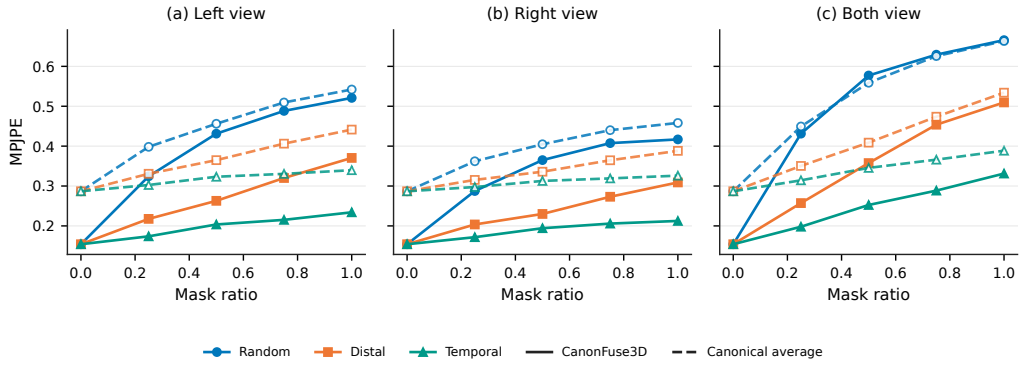


Figure 6: Complete held-out-split masking robustness for left-, right-, and both-view corruption.

5.6. Image-level occlusion through the pose front end

Figure 7 illustrates the three image-level occlusion patterns. Square patches filled with the image mean color are placed on the projected joints selected by the same random, distal, and temporal rules as the pose-level study, with a side length of 12% of the projected person height; the masked frames are then processed by SAM3D and the frozen fusion model, so the resulting errors include missed detections and distorted poses produced by the front end. Supplementary Table S5 lists all 18 conditions.

CanonFuse3D improves on canonical averaging in 15 of the 18 conditions. At random ratio 0.5, gains are 31.9%, 34.3%, and 20.9% for left-, right-, and both-view occlusion, respectively. All twelve distal and temporal conditions retain positive gains, ranging from 14.6% to 43.4% for distal masks and from 42.3% to 45.9% for temporal masks. These results extend the controlled robustness evidence to errors produced by the image-to-pose front end.

Random occlusion at ratio 1.0 reverses the advantage in all three affected-view modes: fused MPJPE exceeds canonical averaging by 18.0%, 9.2%, and 28.4% for left, right, and both views. In the bilateral case, MPJPE is 0.6311 versus 0.4914, and the pair-aligned SAM3D detection-failure rate is 36.36%. Failure rates for the corresponding left- and right-view conditions are

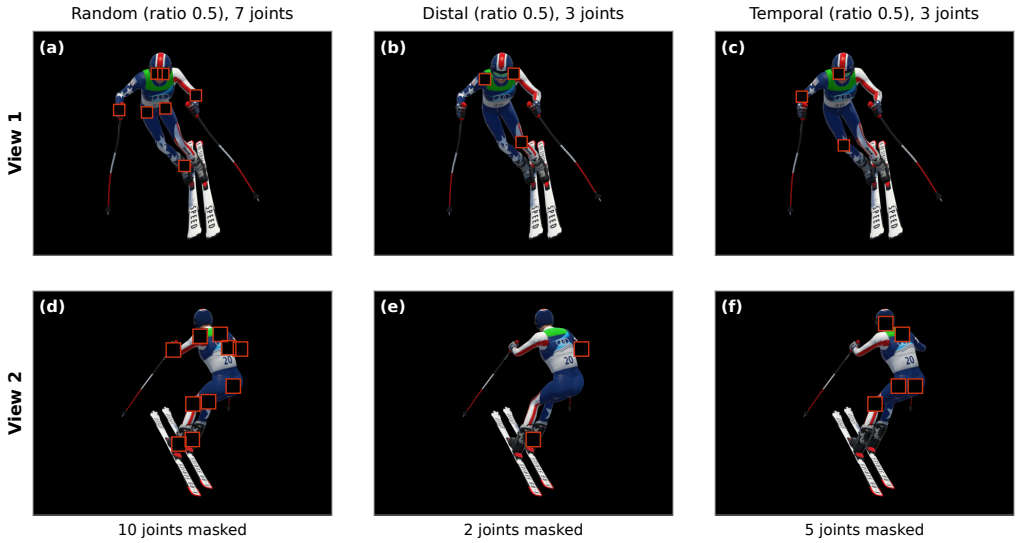


Figure 7: Image-level occlusion examples on one Unity frame. Rows show the two input views and columns the three patterns at ratio 0.5: random joint-frame selection, distal-subset selection, and temporal selection of contiguous 10-frame intervals. Red outlines mark the mean-color patches placed on the selected projected joints; the number of masked joints in each panel is stated in its label. The outlines are drawn for illustration only and are not present in the images passed to SAM3D.

40.51% and 32.22%. Failed detections remain in the evaluation as zero poses, so these results include complete upstream detection loss. The aggregate measurements do not isolate how much of the degradation comes from failed detections, errors in detected poses, or the fusion response. An unoccluded complementary view therefore does not guarantee a fusion advantage under this severe intervention.

Figure 8 compares fused MPJPE for image-level and direct pose-stream interventions. Their matching ratios control different perturbations. For example, bilateral random corruption at ratio 0.5 retains a 20.9% gain in the image-level experiment but gives a 3.3% disadvantage in the pose-level experiment (Supplementary Tables S5 and S4). This difference shows that the two protocols probe distinct error distributions; it does not establish an ordering of physical occlusion severity. The condition counts describe one frozen checkpoint, rather than independent training repetitions.

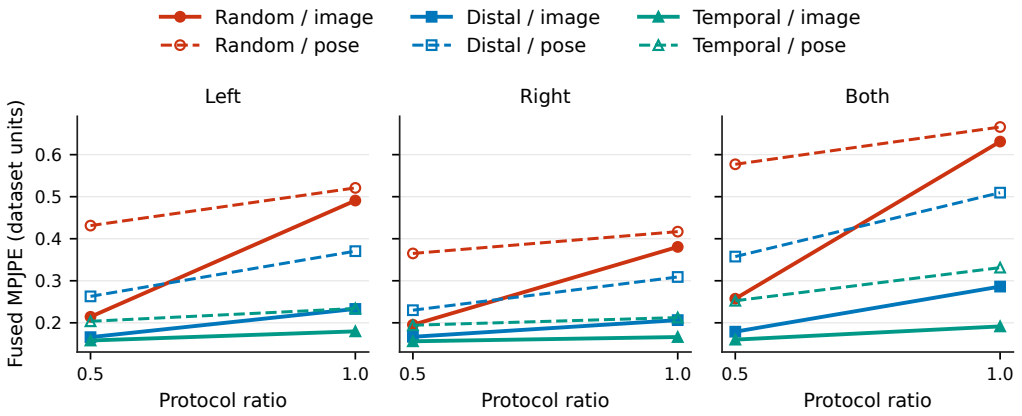


Figure 8: Fused MPJPE under image-level occlusion propagated through SAM3D and direct pose-stream corruption, for left-, right-, and both-view interventions. Solid lines with filled markers denote image-level masks; dashed lines with open markers denote pose-stream corruption. Matching protocol ratios do not imply equal physical occlusion severity. Errors use dataset coordinate units.

5.7. Matched occlusion and calibration sensitivity

We compare pose-only fusion with geometry-assisted reconstruction using predictions from the same corrupted images. One SAM3D call supplies both the original-image 2D keypoints and the 3D pose of the same selected person. These 2D observations are model-projected estimates, not ground truth or outputs of an independent detector. Unity projections locate the occluding patches only. The matched study covers clean inputs and random masks at ratios 0.5 and 1.0 on the left, right, or both views. All five methods retain the same 64,440 windows, batch size 256, sample order, and common-13 scoring after all-15 canonicalization. Learned weights are frozen. This seven-condition study uses fresh paired predictions and a different reported joint subset from the preceding all-15, 18-condition study; their numbers are not interchangeable. Complete protocols and metrics are in Supplementary Section S14.

With unperturbed supplied calibration, DLT-residual has the lowest MPJPE in six of seven conditions. Its clean error is 0.0988 versus 0.1553 for CanonFuse3D. Under the three ratio-0.5 masks, its MPJPE is 25.6–28.8% lower, whereas CanonFuse3D has 1.8–9.5% lower PA-MPJPE and 43.7–46.0% lower acceleration error. Thus, the additional-input method improves position accuracy, while the pose-only method retains advantages

Table 7: Matched RGB-occlusion comparison on Unity, with supplied calibration and no added extrinsic perturbation. Common-13 canonical MPJPE (lower is better), on all 64,440 windows. Ratios select joints for patch placement, not image-area coverage.

Condition	CanonFuse3D	Average	DLT	Gated DLT	DLT-residual
Clean	0.1553	0.2797	0.2004	0.2108	0.0988
Left 0.5	0.2103	0.3051	0.2542	0.2606	0.1498
Right 0.5	0.1949	0.2890	0.2488	0.2556	0.1417
Both 0.5	0.2514	0.3146	0.2957	0.2983	0.1870
Left 1.0	0.4654	0.3992	0.4499	0.4495	0.3569
Right 1.0	0.3706	0.3347	0.5501	0.4569	0.4070
Both 1.0	0.5986	0.4654	0.5121	0.4866	0.4155

CanonFuse3D and canonical averaging use only predicted 3D. The three DLT-based methods additionally use estimated 2D and calibration. DLT follows [28]; gating and the residual MLP are study-specific controls, not reproductions of published learned triangulation systems.

on aligned configuration and temporal second-difference error in these conditions. CanonFuse3D also improves MPJPE over canonical averaging by 20.1–32.5% under ratio-0.5 masks. At ratio 1.0 this advantage reverses for either single-side or bilateral masking, by 10.7–28.6%. The results do not support automatic rejection of an arbitrarily degraded input view.

Figure 9 shows the effect of independently perturbing the supplied right-camera extrinsics by 0°, 1°, and 3°, retaining the left camera and the right-camera center. DLT-residual still has lower MPJPE than CanonFuse3D on clean and ratio-0.5 inputs at 1°. At 3°, CanonFuse3D has the lowest MPJPE and PA-MPJPE among the five methods on those four input conditions, but canonical averaging is best on both metrics in all three ratio-1.0 conditions. Pose-only scores are invariant by construction because neither their images nor their 3D inputs change. This experiment measures sensitivity to supplied calibration, not to physically moving a camera or altering SAM3D's internal camera estimates. Individual conditions need not deteriorate monotonically with angle.

Missing observations remain in all denominators. At bilateral ratio 1.0, at least one view fails in 54.85% of pair–frame positions. After the unchanged batch-referenced transformation, 33.30% of CanonFuse3D and 70.05% of DLT/gated-DLT frames have collapsed common-13 configurations. A missing reference frame can affect an entire batch, rather than only the failed

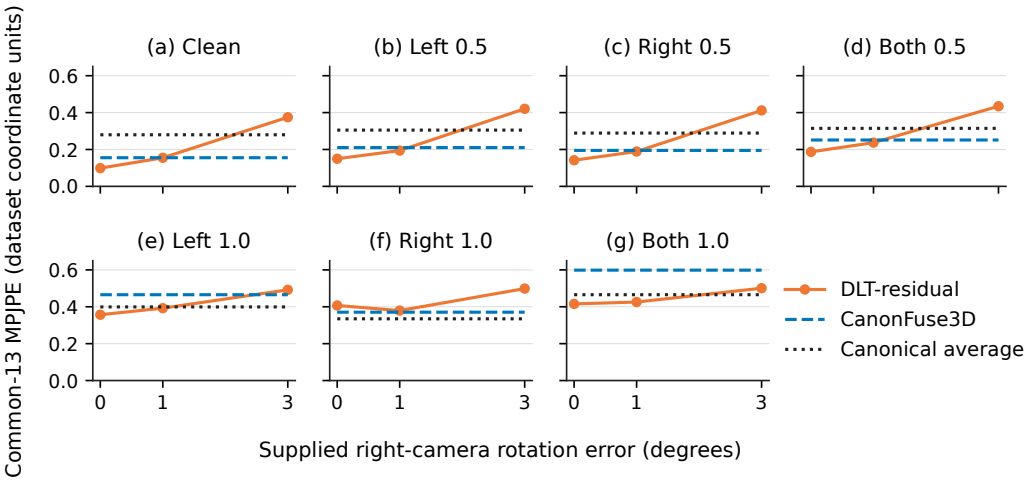


Figure 9: Calibration sensitivity across seven matched conditions. Markers show DLT-residual [28] at the three tested angles; connecting lines are visual guides. Horizontal pose-only references remain unchanged. The synthetic rotation affects only the supplied right-camera orientation; images and camera centers stay fixed. Exact values appear in Supplementary Table S16.

frame. These results characterize the coupled front-end, fallback, normalization, and prediction pipeline; they do not isolate failure of the fusion network or triangulation alone. Small acceleration error for a collapsed output is not evidence of successful motion recovery.

5.8. Front-end generalization and adaptation

CanonFuse3D is trained on SAM3D streams, but the framework itself is front-end agnostic: it consumes only 3D pose sequences on the common joint set and can be placed behind any monocular estimator that provides them. Table 8 first evaluates the frozen SAM3D-trained checkpoint on three alternative lifting front ends without any retraining. MotionBERT retains a positive gain of 8.0% over its own canonical average, whereas PoseFormer and VideoPose3D degrade by 18.0% and 12.8%. Canonicalization removes global position and orientation differences but not estimator-specific limb bias and temporal statistics, so a frozen model remains coupled to the error distribution it was trained on, and transfer to a new estimator should be validated rather than assumed.

Lightweight adaptation restores the benefit on every front end. Table 9 reports the complete 8×4 matrix in which the frozen reference checkpoint

Table 8: Frozen transfer to alternative upstream estimators on the common 13-joint subset. Lower MPJPE is better. Gain is relative to canonical averaging; positive values indicate improvement.

Front end	Canon. avg.	Fused	Gain (%)	Gate pref. (%)
MotionBERT	0.4766	0.4385	+8.0	61.4
PoseFormer	0.4542	0.5360	-18.0	52.3
VideoPose3D	0.4776	0.5389	-12.8	47.0

and seven adapted models are each evaluated on all four front ends; Supplementary Fig. S2 visualizes the corresponding relative recovery. Each adaptation updates either the gate and residual heads only or all parameters for 20 epochs on training and validation actions, without test data. Full target-specific adaptation reduces the matched-front-end error to 0.2290 for VideoPose3D, 0.2741 for PoseFormer, and 0.2393 for MotionBERT, which is 52.0%, 39.7%, and 49.8% below the canonical average of the respective front end (0.4776, 0.4542, and 0.4766) and 57.5%, 48.9%, and 45.4% below the frozen checkpoint on the same input. A single model adapted on a balanced mixture of all four front ends reaches 0.2363, 0.2751, and 0.2464 on the three alternatives while retaining 0.1809 on SAM3D, which is 16.7% above the native checkpoint but still 35.2% below canonical averaging. Thus one architecture, adapted with a short schedule, improves over canonical averaging on every tested front end; the remaining trade-off is between recovery on the target front end and retention on the native one, since no adapted model dominates the frozen checkpoint on every input source.

Table 9: Common-13 MPJPE after front-end-specific or mixed adaptation. Each row is one frozen model evaluated on all four front ends; lower is better.

Adaptation model	SAM3D	VideoPose3D	PoseFormer	MotionBERT
Frozen reference	0.1551	0.5389	0.5360	0.4385
VideoPose3D/heads	0.2120	0.2621	0.3297	0.3567
VideoPose3D/full	0.2442	0.2290	0.2954	0.4403
PoseFormer/heads	0.1822	0.2833	0.3234	0.3521
PoseFormer/full	0.2033	0.2447	0.2741	0.3381
MotionBERT/heads	0.1976	0.4667	0.5100	0.2744
MotionBERT/full	0.2117	0.5050	0.6036	0.2393
Mixed/full	0.1809	0.2363	0.2751	0.2464

5.9. Scaling beyond two views

Figure 10 shows accuracy and bootstrap intervals under frozen pairwise composition, while Table 10 reports its computational cost. Supplementary Table S6 provides the numerical means, standard deviations, and intervals for all methods. Accepted and rejected camera groups are retained in the runtime manifest so that missing common frames are not silently converted into favorable subsets. A separate matched local-model audit (Supplementary Section S11) measures parameter size, matrix/convolution operation counts, and fusion latency for CanonFuse3D, the local TCN, and SmoothNet. CanonFuse3D has 235,524 parameters but higher latency than these two temporal baselines; its purpose is cross-view refinement rather than minimum-cost smoothing. This audit excludes upstream inference and uses a narrower timing boundary than the serial N-view measurements. The method-specific external audit (Supplementary Table S10) measures STRIDE [20], DeciWatch [29], and MetaPose [15] on the same GPU. Their adapted 30-frame paths take 755.513 ms including STRIDE's 30 test-time updates, 3.123 ms for DeciWatch's sparse recovery, and 18.927 ms including MetaPose's initialization and coordinate adapters. STRIDE's final inference alone takes 12.095 ms and must not be substituted for its full adaptation cost. MetaPose retains its FP64 learned solver. Input joints, host transfers, and computational contracts differ, so these values describe deployment costs rather than a matched-input speed ranking; none includes the RGB front end. The two-view configuration captures the largest gain: it reduces MPJPE from the single-view value of 0.1715 to 0.1295 (24.5%) with one pair forward and

Table 10: Accuracy–cost trade-off for deployable frozen pairwise composition. Latency is the serial fusion time per 30-frame camera group; reduction is relative to the two-view MPJPE.

Views	Pairs	MPJPE	Red. (%)	Latency (ms)	Rel.	Groups/s	Peak MiB
2	1	0.1295	–	2.02	1.00×	494.7	12.28
3	3	0.1177	9.1	6.02	2.98×	166.1	12.31
4	6	0.1158	10.6	12.06	5.96×	83.0	12.33

Serial latency is the synchronized mean over 44 accepted 30-frame camera groups after 10 warm-up executions on one NVIDIA RTX A6000. It includes pose canonicalization, all unique pair forwards, and output averaging, but excludes monocular front-end inference, data loading, and oracle selection.

2.02 ms serial fusion latency. Three views reduce MPJPE by a further 9.1% relative to two views, but require three pair forwards and 6.02 ms (2.98-fold) latency. Four views reach 0.1158 MPJPE, a 10.6% reduction relative to two views, while requiring six pair forwards and 12.06 ms (5.96-fold) latency; the incremental gain from three to four views is only 1.6%. Thus, additional views remain useful when accuracy is prioritized, whereas two views provide the most favorable accuracy–cost operating point for the main CanonFuse3D configuration.

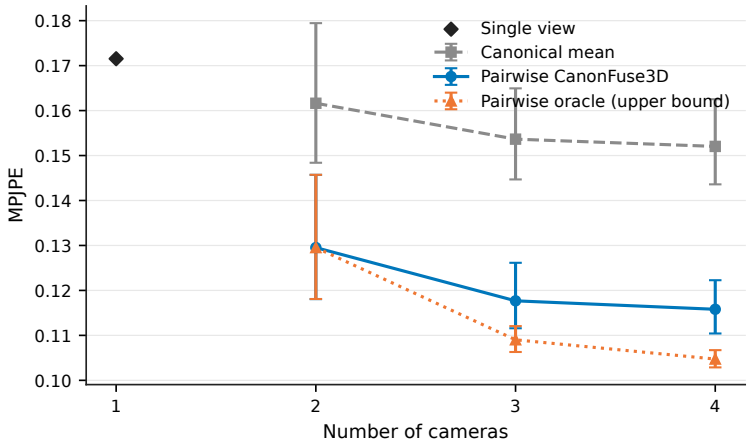


Figure 10: N-view scaling under frozen pairwise CanonFuse3D composition. Error bars show paired-group bootstrap 95% confidence intervals. The oracle is an upper bound.

5.10. Automatic synchronization diagnostic

Figure 11 compares uncorrected, automatically corrected, and oracle streams across the eleven injected offsets. Supplementary Table S7 reports the exact errors together with lag-estimation accuracy and correction-activation rates. The confidence threshold (0.09296) was fixed from validation data before these test results were read. The result is directionally asymmetric. Automatic correction improves MPJPE for every tested positive lag: at +5 frames, it reduces error from 0.1537 to 0.1506, versus the 0.1488 oracle. It also gives smaller improvements at ±0.5 frames. In contrast, it worsens every tested negative integer lead; at −5 frames, error increases from 0.1484 to 0.1498. At zero offset, 41.5% of samples activate a nonzero correction and MPJPE increases slightly from 0.14879 to 0.14884. Exact lag accuracy is only 13.2% at +5 frames. Thus, the descriptor is a useful diagnostic and partially recovers positive delays, but it is not a reliable bidirectional synchronization solution. We do not use it to strengthen the main CanonFuse3D comparison.

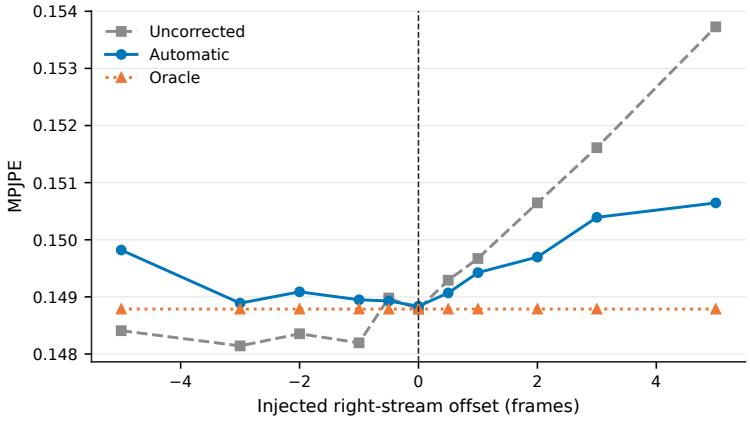


Figure 11: MPJPE under injected right-stream offsets on the fixed central 20-frame interval. Oracle restores the original unperturbed right stream; automatic correction uses the validation-calibrated confidence threshold.

5.11. Qualitative evaluation

Figure 12 illustrates deployment on two author-collected clips not shown in the earlier qualitative figure. Left- and right-view poses are estimated independently, and the frozen Unity-trained checkpoint processes a 30-frame window without camera calibration. Fixed midpoint selection avoids choosing

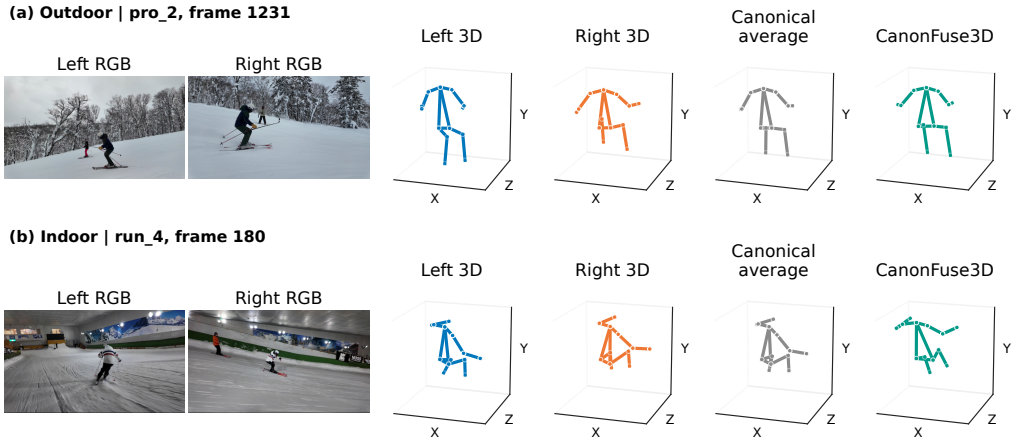


Figure 12: New qualitative examples from author-collected (a) outdoor and (b) indoor recordings. Each row shows the two full input frames, the canonicalized single-view predictions, their average, and the frozen Unity-trained CanonFuse3D output. Examples use the midpoint of the shared frame index in two previously unshown clips, fixed before inference, with a 30-frame input window. All 3D panels show the same 13 body joints at a shared orientation and scale, without synthesized landmarks. These recordings have no independent metric 3D ground truth and are excluded from training and quantitative evaluation.

examples by apparent reconstruction quality; separate panels expose cross-view disagreement and the changes introduced by averaging and learned refinement. The shared display scale preserves differences between the outputs. Without independent 3D ground truth, these examples demonstrate the inference workflow, not an accuracy ranking. Similar errors in both upstream predictions can remain in the refined output.

6. Discussion

CanonFuse3D's advantage over non-learned pose-only controls supports learned refinement of complementary 3D streams, but the benchmark identifies no universal winner (Tables 2 and 3). Learned temporal methods trade position accuracy against smoothness, while geometric methods require additional 2D observations and calibration. The matched Unity ablation supports contributions from both gating and output residual correction to position accuracy, with an acceleration-error trade-off. This remains single-seed evidence and does not include Ski-PTZ-Pose retraining. The native-output

audit and canonical-space comparison answer different questions and must not be combined into one ranking.

The useful deployment regime is approximately synchronized input with complementary evidence and an error distribution represented during training. Tolerance to the tested camera separations and small timing perturbations does not establish robustness to long delays, dropped frames, or independently varying frame rates. Pairwise composition extends the frozen model to additional cameras without learning a joint multi-view representation. Its serial cost grows with the number of pairs, making two views the main accuracy–cost operating point in the evaluated implementation. Batching or parallelizing independent pairs could reduce latency without changing that pair count.

Estimator-agnostic architecture does not imply estimator-independent weights. Target-specific and mixed adaptation restore gains across the tested front ends, but recovery on a new estimator trades off against retention on SAM3D. A new deployment should therefore validate its upstream error distribution and, where needed, adapt the fusion model. The alternative-front-end experiments isolate 3D lifting with ground-truth Unity 2D tracks; they do not establish transfer under image-detection noise.

The RGB studies reveal limits that pose-only perturbations cannot establish. An unoccluded second view does not guarantee improvement over averaging under severe masking. Accurate supplied calibration can favor learned geometric correction, while larger tested calibration errors favor pose-only fusion only in the clean and moderately masked conditions (Section 5.7). Position, aligned-shape, and acceleration errors need not favor the same method: improved temporal agreement does not ensure accurate reconstruction, particularly when outputs collapse. The matched-image design reduces one source of comparison ambiguity, but geometric controls still use additional observations and calibration. It characterizes sensitivity to supplied geometry, not the cost or accuracy of estimating that geometry during deployment.

Artificial patches and one frozen SAM3D front end do not establish robustness to natural occluders, weather, or motion blur. Blur may create temporally correlated localization or detection errors and disturb anchor landmarks, but these mechanisms were not measured here. The association between detection failure and fusion degradation likewise does not isolate a causal mechanism.

The reference regimes also limit interpretation. Unity provides an avatar

skeleton in dataset coordinates, not a metric motion-capture measurement. Ski-PTZ-Pose errors measure agreement with the constructed reference derived from aligned view-specific annotations, not independent physical 3D accuracy. Raw and transformed errors are not interchangeable, skeleton scale is not normalized, and the reported values should not be relabeled as millimeters.

The evaluated body reference is shared within each batch. Changing batch composition can therefore change the representation, and missing reference landmarks can collapse other frames in that batch (Supplementary Section S14). These results neither establish invariance to regrouping or single-sequence deployment nor isolate the prediction module's contribution to severe-corruption failures. A sample-independent variant requires retraining and evaluation. Supplementary Section S9 records the exact reference convention.

Finally, the gate is a learned reliability signal, not calibrated uncertainty: balance and entropy regularization shape its mean, and error correlation or preference accuracy does not establish probabilistic calibration. Frozen-checkpoint stress tests do not measure variation across training runs. Sample- or group-bootstrap intervals describe resampling uncertainty conditional on the checkpoint and sampling units, not variability across seeds or folds. Between-dataset uncertainty remains unresolved. Broader validation requires external sports data, independent metric references, longer asynchronous sequences, and naturally occurring image degradation.

7. Conclusion

CanonFuse3D refines two approximately synchronized monocular 3D streams without images, 2D observations, or camera parameters at the fusion stage. It reduces canonical-space position error relative to averaging by 44.5% on Unity and 48.6% on Ski-PTZ-Pose. Learned temporal and geometry-assisted methods retain advantages on particular metrics or input regimes. Controlled studies delimit tolerance to camera separation, timing errors, and corruption: accurate supplied calibration can favor geometric correction, whereas larger tested calibration errors favor pose-only fusion on clean and moderately masked inputs. Severe image masking remains a failure regime, even with one unmasked view. Adaptation supports alternative front ends, and pairwise composition supports additional cameras, with two views offering the main evaluated accuracy–cost operating point. These

fixed-partition, single-run results support deployment only after validating the upstream estimator and capture conditions. Cross-seed variability, independent physical 3D accuracy on real recordings, and broader real-world generalization remain open.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Kaixu Chen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Qingyu Wang:** Software, Investigation, Data curation, Validation. **Chun Xie:** Methodology, Resources, Supervision, Funding acquisition, Writing – review & editing. **Eiichi Naito:** Conceptualization, Resources, Writing – review & editing. **Toshitaka Kimura:** Conceptualization, Resources, Writing – review & editing. **Itaru Kitahara:** Conceptualization, Resources, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Data and code availability

The Ski-PTZ-Pose dataset is publicly available from the dataset provider [25]. The source code of CanonFuse3D, the training and evaluation scripts, the fixed train–validation–test indices, and the evaluation manifests recording checkpoint and front-end estimator hashes are available at `https://github.com/ChenKaiXuSan/skiing_dual2pose`. The Unity synthetic dataset and the frozen CanonFuse3D checkpoint used in all reported experiments will be deposited in a public repository with a persistent identifier upon publication. The author-collected indoor and outdoor recordings contain identifiable individuals and are not shared; they are used solely for qualitative illustration and contribute neither training samples nor quantitative measurements.

Ethics statement

The quantitative experiments use a synthetic dataset and the previously released Ski-PTZ-Pose dataset and involved no new data collection from human participants. The author-collected indoor and outdoor recordings used for qualitative illustration show adult volunteers from the authors' group who gave written informed consent to be recorded and to appear in the published figures. No other identifiable individuals are shown, and no personal data beyond the illustrated frames are released. The recordings were non-invasive observational video captures of consenting adults and did not require institutional ethics review under the guidelines of the University of Tsukuba.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) and OpenAI Codex in order to improve the readability and language of the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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